

Project Report (CSCE 5214)

Group 4 – Spaceship Titanic

Executive Summary

In this project we have developed a user interactive website. The website takes input from users about records of a passenger and gives the choice to the user to choose between two machine learning models which would predict and return whether the passenger has been transported to an alternate dimension or not due to the space-time anomaly in which the spaceship was trapped. The scope of this project is vast as it can be applied to any transport system scenarios. Here, we have learned about the correlation of various features and applied that to get a better perspective of the data. Using that perspective, we can use the similar intuition to other real-life problems like survival of passengers after a bus accident or a plane crash. This solution can be extended and be used in business scenarios where we can use this program to predict whether a customer is likely to buy tickets of seat in a bus or train or plane or even in a movie theatre or game show based on the personal records of the customer and previous data showing how likely a customer is to buy that ticket.

1. Introduction

Our project is titled as the “**Spaceship Titanic**”. It is taken from a Kaggle competition. In this problem, there is a spaceship named Titanic, which is on a voyage to transport passengers to another habitable planet. But during its voyage, it is met with a space-time anomaly due to which, almost half of the passengers were transported to an alternate dimension. Using the records from the computer system of the spaceship, we have to design a model, train it and predict which passengers are transported to the alternate dimension.

In this project, we have fairly distributed data across both the classes, which are transported to the alternate dimension (denoted by the target label “True”) and which are not transported (denoted by the target label “False”). There are quite a few missing data in the samples which are handled by us using KNN imputer and mean values of features. We have used target encoding and label encoding to change the categorical values into numerical values.

In our proposed solution we have used two machine learning models – CatBoost and Logistic Regression – to classify the data samples and to predict the test samples.

Spaceship Titanic ([Link to the Kaggle competition](#))

2. Architecture

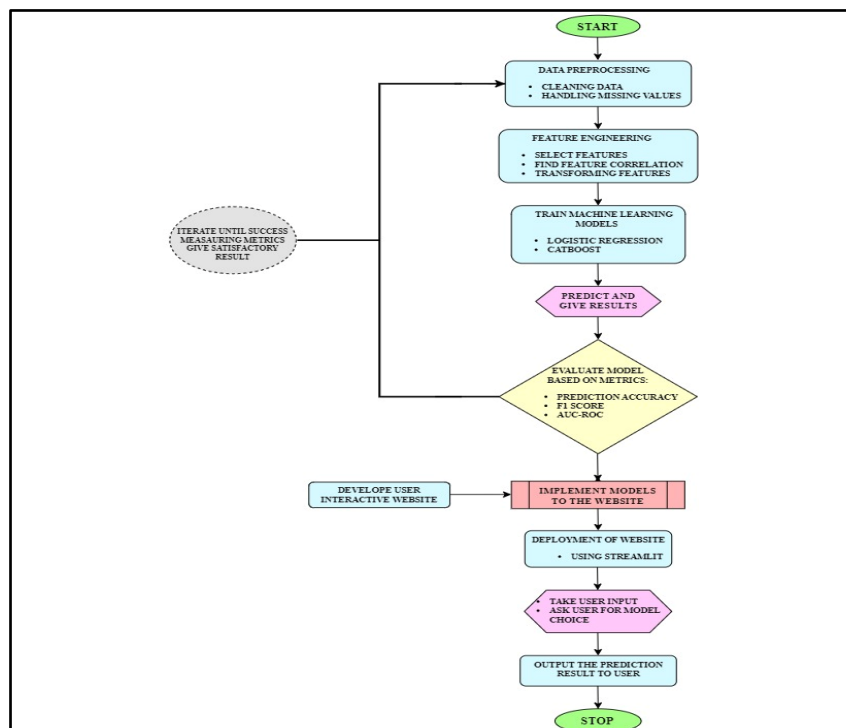


Fig.1 Flowchart of all the phases involved in the Project

3. Modelling

Link to the Dataset - <https://www.kaggle.com/competitions/spaceship-titanic/data>

The Spaceship Titanic dataset has 14 columns, including the target variable column. The target variable is to predict whether a particular passenger has been transported. Among the rest of the 13 columns, Home Planet, cryosleep, Destination, VIP and Name are the categorical data and the cabin column combines categorical and numerical data. The columns, Room Service, Food Court, Shopping Mall, Spa, and VR Deck represent the expenses made by passengers in the respective categories.

The data pre-processing began by observing the data and making a few changes to the columns in the given dataset. The first observation was that the 5 columns representing various expenses were not individually affecting the target variable at a significant rate, hence they were added together to form a single column representing the 'Total Expenses' of the passengers. Next, we observed that the Cabin column consisted of three parts: Deck, Number, and side. When observed we noticed significant relations between Deck and Homeplanet. Hence, the Cabin column has been split into three columns to represent its three parts.

Now that the required changes with the columns have been done, we went ahead with the process of handling the missing values. There were certain observations made within the data to handle a part of the missing values. The observations made and the corresponding changes are listed below:

1. If passengers are in cryosleep then the total expenses will be zero.
2. If the passengers don't have total expenses equal to zero then they can't be in cryosleep.
3. Passengers under the age of 18 can't be VIP members.
4. There are no VIPs from HomePlanet VIP.
5. People from the same group in the passenger ID are from the same HomePlanet.
6. link between HomePlanet and CabinDeck:
Earth on E, F, G
Mars on D, E, F
Europa on A, B, C, D, E, T

Using these logics a few of the missing values in HomePlanet have been substituted.

After inputting the missing values using these relations, the rest of the missing values have been imputed using a KNN imputer as it preserves the underlying patterns in the data to maintain the accuracy of the model.

We have chosen two models for this particular project, Catboost and Logistic Regression. Catboost, with its ability to capture the complex relations within the data even with limited pre-processing, makes it a good choice to use for this binary classification. Whereas Logistic Regression is a simpler approach but gives insights into factors affecting the target variable.

The total training dataset was split into train and validation dataset. For the Logistic Regression model, the train test split was done to 70-30 at first, then 75-25 and finally 80-20. And then, throughout this project we have finalized the train test split of the training dataset as 80-20. The Logistic Regression model is fit and trained over the 4 versions of the preprocessed and feature engineered dataset. In the attempt 4 it is giving the overall best result of Accuracy, F1 score and AUC-ROC. So that model is chosen for the project. For the case of CatBoost, in Attempt 1 of hyperparameter tuning, the `n_neighbors` for the KNNImputer was set to 5, striking a balance between data sensitivity and noise reduction. For the CatBoostClassifier, a verbose level of 100 was specified to monitor training progress, while `cat_features` was used to identify categorical variables for optimized handling. These adjustments aimed to enhance the model's accuracy and robustness. In Attempt 2, various hyperparameters were tuned, including `learning_rate`, `depth`, `l2_leaf_reg`, `iterations`, and `early_stopping_rounds`. The `learning_rate` controls the model's learning speed, while `depth` determines the complexity of the trees, and `l2_leaf_reg` helps prevent overfitting. In Attempt 3, the same hyperparameters were fine-tuned again, highlighting their critical role in balancing model complexity and performance. Effective tuning of these hyperparameters is essential for improving predictive accuracy by preventing both underfitting and overfitting.

Test Results:

The Table.1 shows a very good representation of the performance metrics achieved through both the models.

Model Name	Attempt Number	Accuracy	F1 Score	AUC-ROC
Logistic Regression	1	61.36%	0.6098	0.6669
	2	59.23%	0.5599	0.6478
	3	57.96%	0.5657	0.6249
	4	62.22%	0.6156	0.6748
CatBoost	1	75.27%	0.7422	0.8271
	2	78.70%	0.7984	0.8751
	3	79.79%	0.7914	0.8611

Table.1 Performance measuring metrics of both the models used

4. Implementation

The project will be implemented locally using the streamlit framework for a desktop website.

We can install streamlit using pip command as follows:

1. **pip install streamlit** (to install streamlit)
2. **streamlit run deployment.py** (assuming deployment.py is our final code)

Load Models:

- The trained logistic regression model with preprocessed train data is loaded using joblib.
- The trained CatBoost model is also loaded in the same way.

File Upload:

- Users will get an option to upload a CSV file for prediction.
- If no file is uploaded, the app loads pre-existing file train.csv by default.

Input:

- The app takes the PassengerId as input for prediction avoiding using multiple input fields.

Prediction Button:

- A "Predict" button triggers the prediction process calling API from trained models.

Data Fetching:

- Based on the entered PassengerId, the corresponding passenger data is fetched from the loaded dataset.

Preprocessing:

- The passenger's data is transformed using the saved preprocessing technique from the logistic regression model and CatBoost model.

Prediction:

- Predictions are made using both the logistic regression model and the CatBoost model.
- The final prediction is calculated as the average of both models' predictions, and the result is displayed as "Yes" or "No" for whether the passenger was transported.

Download Option:

- The deployment code provides an option to download the prediction results as a CSV file.

Handling Missing Passenger ID:

- If the entered PassengerId does not exist in the dataset, an error message is shown.

Github Link to the Repository - <https://github.com/4sameek4/SpaceshipTitanic.git>

5. Conclusion

The first target of our project was to finish the project 3-5 days before the submission deadline, so that we could do more tuning of the models and some final beautification to the designing of the website. We also aimed to achieve a fairly considerable prediction accuracy of the Logistic Regression model to be around 70% and F1 score and AUC-ROC values accordingly, which we could reach up to 62.22% due to time constraints. The project successfully used Logistic Regression for its ease of use and clear interpretation, with the best results showing 62.22% accuracy, an F1 score of 0.6156, and an AUC-ROC of 0.6748 after refining the dataset. Tuning hyperparameters for both Logistic Regression and CatBoost helped improve performance, especially in the CatBoost model by adjusting factors like learning rate and depth. Nevertheless, Logistic Regression's assumption of a simple linear relationship limited its accuracy. Future improvements could involve trying more advanced models, fine-tuning parameters further, and testing different data preprocessing methods. Tools like SHAP (SHapley Additive exPlanations) could also help explain which features matter most.

6. Individual Contribution

- **Sameek Bhattacharya (UNT ID : 11776897) – Project Manager**

I have served as a contribution to this project from the managerial as well as technical point of view. After getting the preprocessed training data samples, I have changed the categorical values into numerical values using label encoding (that is, True values to 1 and False values as 0) and target encoding (that is using the target encoder to encode values having 2 or more than 2 variations with their mean values. Like the feature deck having 6 different values and home planet and destination planet having 3 different values.)

After having the final dataset, I have split the overall training data samples into 80% of it as training set and 20% of it as validation set. Then I used the Logistic Regression model from the sci-kit learn package. I have trained the model on the training set and used it over the validation set for measuring the success of prediction. The metrics used for measuring were Classification Prediction Accuracy, F1 score and AUC-ROC. I have used this model over 4 different versions of the feature engineered and preprocessed dataset. The highest success coming on the 4th attempt of Accuracy 62.22%, F1 score 0.6156 and AUC-ROC 0.6748.

As a project manager, I have integrated all the information gathered from individual team members and documented them in a generalized format. From this project I have learnt how to manage a team of 4 members each having a different set of skills and time schedule throughout the week. I have learnt how to setup up milestones in the project and deadlines for each milestone. I have also learnt the value of responsibility and coordination as an individual for the smooth and successful deployment of the project.

- **Sharmila Eday (UNT ID : 11816083)**

My contribution to the project focuses on data analysis, finding the data relations, and feature engineering aspects of the project. Data analysis started with checking whether the data was balanced to advise on the metrics that we would need. After checking the class balances, I have concluded that the dataset for this particular project is balanced. Moving ahead I plotted the individual columns concerning the target variable to analyze the effect of each feature on the target variable. From these plots, I concluded that cryosleep is an important factor i.e. the people in cryosleep are more likely to get transported. Another conclusion was that the five expenses columns didn't have a significant difference in their impact on the transported column, hence I decided to combine them to form the 'Total Expenses' column.

In the feature engineering part, the major part of the work included recognizing the relations between the various features given in the dataset. Identifying the patterns and correlations among data helps in filling the missing values. Through my observations, I have noticed the relations among the columns cryosleep and Total Expenses, Age and VIP status, Home Planet and VIP status, Home Planet, and the Decks of the Cabin. Then I have filled in the missing values according to the relations found which have been discussed in detail in the Modelling section of this report. One more logic that I have applied is that the passenger ID consists of the group number as well as the person number but as the whole

group boards from the same Home Planet, any missing entries for the Home Planet can be filled in based on the other passengers' Home Planet within that group.

After using these relations, certain missing values remain which I have filled with the KNN imputation method. KNN imputation was the method that stood out among the other possible methods due to its ability to consider the underlying patterns in the dataset which will help the overall accuracy when this dataset is used for the models.

For the further scope of the feature engineering part, I looked into the numeric transformations of the data to enhance it further. Normalization and log transformations are the two methods that I have checked on the dataset. Normalization ensures that the data is scaled consistently which is useful in gradient boosting methods as it relies on distance calculations. It also helps in faster convergence in the gradient boosting methods. Whereas the log transformations handle the skewness of the data by controlling the impact of the outliers in the data. Together these processes would have improved the quality and robustness of the model. However, these haven't been applied to the final dataset considered for the models as of now due to the time constraints.

- **Invika Reddy Balireddi Gari (UNT ID : 11712523)**

My contribution in this project was applying the CatBoost model to improve prediction accuracy. In the second attempt at tuning, the model achieved impressive results, with an accuracy of 78.70%, an F1 score of 0.7984, and an AUC-ROC of 0.8751. These outcomes highlighted the effectiveness of the hyperparameter adjustments made during this phase. The improved results indicated a better balance between identifying true positives and reducing false positives. Overall, it was satisfying to see such significant progress in the model's performance.

I have actively attended and participated in group discussions and made a consistent progress to achieve the desired output. The regular check-ins and clear communication among the team members have helped me in aligning goals and addressing any issues promptly. Being flexible and willing to coordinate has lead to valuable insights regarding the designing of my CatBoost model.

- **Aashish Pandey (UNT ID : 11761533)**

In this project, I am tasked with implementation of the trained logistic regression and CatBoost model for predicting whether the passenger was transported or not. It's a simple web app hosted locally using the streamlit python framework.

For my part, I wrote code to accept a csv file or PassengerId for prediction and added a predict button in the original UI. After checking if the user was transported or not, we will get the option to download the result in csv format for analysis.

Acknowledgement:

This project report is documented and verified by all the team members Sameek Bhattacharya, Sharmila Eday, Invika Reddy Balireddi Gari and Aashish Pandey. The section of "Individual Contribution" is the documentation of each of the team members submitted separately as per their own experiences and opinions. It is just pasted in this document directly only making changes for complying to the overall font, alignment and orientation format of the document.